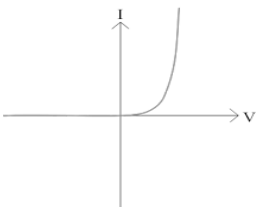


| Question |     |      |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks Available |          |          |           |          |           |
|----------|-----|------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |     |      |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
|          | (d) | (i)  |  | <p>Connect in series one way <b>and</b> see if lamp lights <b>or</b> the resistance will be low <b>or</b> current will be high (1)<br/>           Reverse the cell / battery / + and - / connection / box (1)<br/>           See if the lamp still lights <b>or</b> the resistance will be much higher <b>or</b> diode only lets current flow one way <b>or</b> current will be zero (very small) (1)</p> <p><b>Alternative:</b><br/>           Replace lamp with sealed box / add in series [with lamp] (1)<br/>           Vary <math>R</math> and take series of reading of current and voltage (1)<br/>           Reverse box / polarity of cell and repeat step 2 (1)</p> |                 |          | 3        | 3         |          | 3         |
|          |     | (ii) |  |  <p>Don't accept S shapes or curve with a decreasing gradient</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1               |          |          | 1         |          | 1         |
|          |     |      |  | <b>Question 6 total</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>4</b>        | <b>5</b> | <b>6</b> | <b>15</b> | <b>6</b> | <b>15</b> |